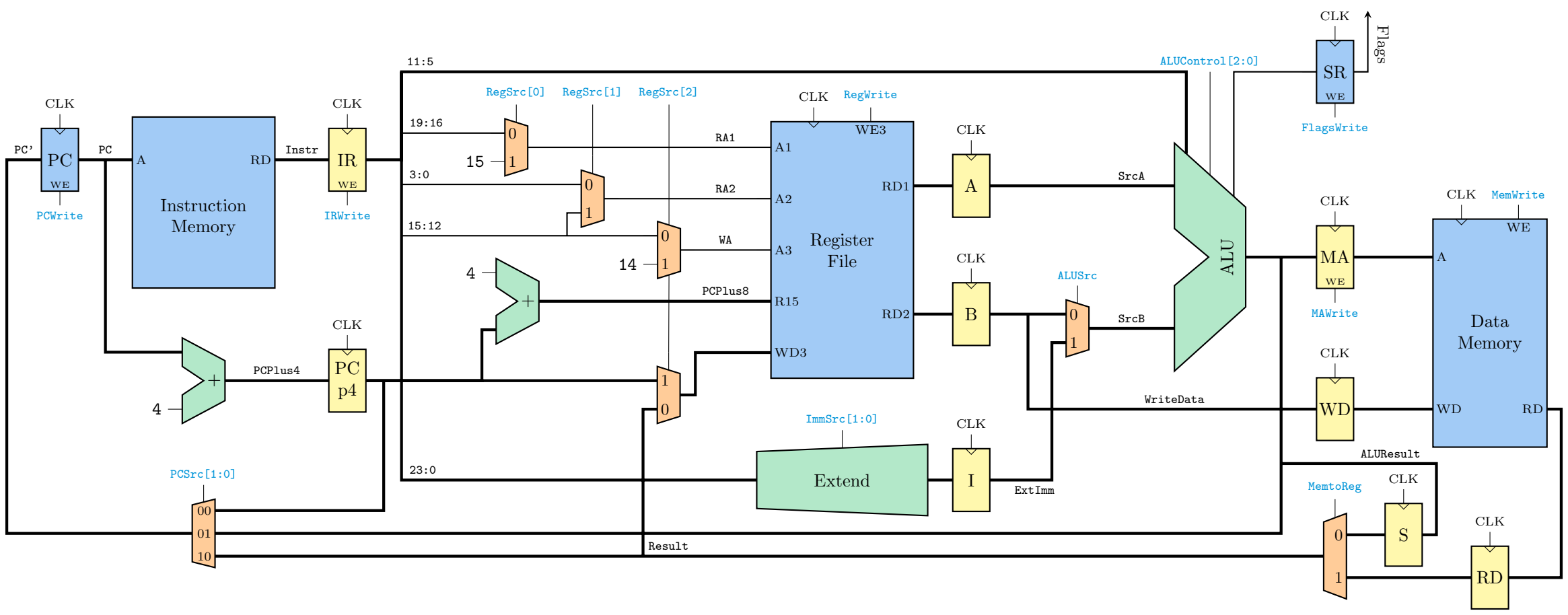

CLK
PC'
PC
WE
PCWrite
PC
A
RD
Instruction Memory
Instr
CLK
IR
WE
IRWrite
11:5
19:16
15
RegSrc[0]
0
1
3:0
RegSrc[1]
15:12
14
RegSrc[2]
4
+
PCPlus8
CLK
RegWrite
WE3
A1
A2
A3
R15
WD3
RA1
RA2
WA
RD1
RD2
Register File
CLK
A
CLK
B
ALUSrc
SrcA
SrcB
ALU
ALUControl[2:0]
CLK
SR
WE
FlagsWrite
Flags
CLK
MA
WE
MAWrite
CLK
MemWrite
WE
A
Data Memory
WD
RD
WriteData
CLK
WD
ALUResult
4
+
PCPlus4
CLK
PC p4
PCSrc[1:0]
00
01
10
23:0
ImmSrc[1:0]
Extend
CLK
I
ExtImm
Result
MemtoReg
CLK
S
CLK
RD